

ARI5118 — Deep Learning for Computer Vision

Deliverable 2: Adversarial Quiz Technical Rationale

Topic 5: Depthwise Separable Convolutions & Efficient Architectures

1 Introduction and Architectural Objectives

This document establishes the underlying software engineering, computer vision, and mathematical rationales for the design of the accompanying 10-question Adversarial Quiz. The quiz is constructed to explicitly counter surface-level rote memorisation by forcing candidates to demonstrate precise structural clarity regarding spatial arithmetic, dimensional mapping, and boundary conditions within mobile-optimised topologies.

2 Itemised Technical Analysis & Distractor Rationales

Question 1: Parameter Count Scaling Efficiency

Core Concept: Parameter reduction constraints of Depthwise Separable layers.

Analytical Verification: The parameter cost reduction ratio of a Depthwise Separable Convolution relative to a Standard Convolution is expressed analytically as:

$$\text{Cost Ratio} = \frac{1}{N} + \frac{1}{D_k^2}$$

As the target output channel depth $N \rightarrow \infty$, the limit of the first fractional term $\frac{1}{N}$ vanishes to zero. The remaining asymptotic lower bound is completely governed by the spatial surface dimensions of the depthwise matrix, yielding an invariant value of $\frac{1}{D_k^2}$.

Distractor Pitfalls: Candidates frequently assume N scales both configurations identically, failing to observe that the independent depthwise filtering phase does not process or interact with N .

Question 2: Receptive Field Calculation Stacking

Core Concept: Effective Receptive Field (ERF) propagation tracking.

Analytical Verification: The cumulative spatial span of consecutive convolution operations scales additively according to the discrete recurrence relation:

$$\text{RF}_{\text{out}} = \text{RF}_{\text{in}} + (K_{\text{kernel}} - 1) \times S_{\text{previous}}$$

Layer 1 (3x3 depthwise filter) establishes a spatial foundation of 3. Layer 2 (5x5 kernel, stride=1) appends $(5 - 1) \times 1 = 4$ units to the spatial footprint, resulting in an ERF of $3 + 4 = 7 \times 7$. The intermediate 1x1 pointwise layer does not alter spatial span.

Distractor Pitfalls: Students routinely multiply kernel lengths ($3 \times 5 = 15$), which violates the overlapping linear stacking laws of spatial receptive fields.

Question 3: Pointwise Convolution Output Map Dimensions

Core Concept: Explicit application of spatial convolution arithmetic on 1x1 filters.

Analytical Verification: Discrete dimension reduction maps uniformly across all kernel sizes:

$$W_{\text{out}} = \left\lfloor \frac{W_{\text{in}} - K_{\text{kernel}} + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{14 - 1 + 2(1)}{2} \right\rfloor + 1 = 7 + 1 = 8$$

Distractor Pitfalls: A common misconception dictates that 1x1 pointwise layers cannot natively support padding configurations or non-unit strides, leading to an incorrect prediction of 7x7.

Question 4: MobileNetV2 Activation Layer Removal

Core Concept: Information manifold retention within compressed vector representations.

Analytical Verification: Inverted bottleneck layers compress high-dimensional representations down into low-dimensional bottlenecks. Applying a non-linear activation threshold (such as ReLU6) directly to narrow channel manifolds forces negative outputs to zero, causing irreversible information collapse. Maintaining a linear projection layer preserves structural features.

Distractor Pitfalls: Candidates confuse generalised convolutional conventions with the explicit information-theoretic exceptions required by narrow bottleneck layers.

Question 5: Stride Impact on Skip Connections

Core Concept: Topological compatibility rules of identity residual mapping.

Analytical Verification: Element-wise matrix addition ($\mathbf{x} + \mathcal{F}(\mathbf{x})$) strictly requires identical spatial ($H \times W$) and depth (C) tensor volumes. Enforcing a stride of 2 collapses the spatial grid dimensions during the depthwise phase, creating an architectural shape mismatch that blocks identity shortcuts.

Distractor Pitfalls: Students often assume residual pipelines integrate auto-pooling mechanisms, overlooking that MobileNetV2 shortcuts must remain pure identity paths to minimise hardware latency.

Question 6: Depthwise Layer Cross-Channel Interleaving

Core Concept: Isolation of spatial calculations in independent channel maps.

Analytical Verification: The depthwise step applies exactly one 2D spatial mask to each channel slice independently with zero cross-channel mixing. Computing a single localised pixel output within a channel maps directly to the spatial matrix footprint, requiring exactly D_k^2 operations.

Distractor Pitfalls: Candidates mistake this phase for a standard convolution operation, scaling the localised arithmetic count by multiplying it by the total input depth M .

Question 7: Theoretical MFLOPs Calculation

Core Concept: Quantitative profiling of Multiply-Accumulate execution loops.

Analytical Verification:

$$\text{Standard Cost} = 56 \times 56 \times 3 \times 3 \times 64 \times 128 = 231.21 \times 10^6 \text{ ops}$$

$$\text{Separable Cost} = (56 \times 56 \times 3 \times 3 \times 64) + (56 \times 56 \times 1 \times 1 \times 64 \times 128) = 27.50 \times 10^6 \text{ ops}$$

The specific variance across independent phases yields a fractional remainder value of 22.42 MFLOPs.

Distractor Pitfalls: Inaccuracies stem from omitting the independent depthwise calculation cost or improperly accumulating the input-to-output channel interactions.

Question 8: The "Inverted" Residual Topology

Core Concept: Structural layer configurations between classic ResNet and MobileNetV2.

Analytical Verification: Classic ResNet routing links high-dimensional tensor matrices via shortcuts, compressing them internally via bottlenecks. MobileNetV2 inverts this paradigm by structuring skip connections specifically across thin bottleneck layers, expanding features internally to a high-dimensional space for spatial filtering.

Distractor Pitfalls: Students swap the architectural ordering, failing to track how channel depth changes throughout the block.

Question 9: Dilation vs Stride Arithmetic

Core Concept: Computing effective spatial receptive spans under dilated configurations.

Analytical Verification: The effective spatial size of a kernel expanded by a dilation factor r is:

$$K_{\text{eff}} = D_k + (D_k - 1)(r - 1) = 3 + (3 - 1)(2 - 1) = 5$$

Evaluating the inverse dimension mapping with zero padding gives: $W_{\text{in}} = W_{\text{out}} + K_{\text{eff}} - 1 = 6 + 5 - 1 = 10$.

Distractor Pitfalls: Candidates apply the base un-dilated kernel size of 3, leading to an incorrect input size of 8.

Question 10: Memory Footprint Constraints

Core Concept: Distinguishing weight parameter savings from training activation cache space.

Analytical Verification: Depthwise separable layers minimise parameter storage, but back-propagation optimisation loops require keeping the intermediary activation tensor generated between the independent depthwise and pointwise operations in cache memory, leading to an increased training footprint.

Distractor Pitfalls: A common misconception is that a drop in parameter weights guarantees an identical, linear reduction in memory usage during training.